THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** If the diagonals of a quadrilateral bisect each other, then it is a rhombus.	swap the 'if' part and the 'then' part
2	**(a)** 9	it satisfies the 'if' part but not the 'then' part
3	**(d)** false; 8 is a multiple of 4 but not of 12	8 = 4×2 is a counterexample
4	**(c)** right-angled, since $a^2 + b^2 = c^2$	$49 + 576 = 625$ and $25^2 = 625$
5	**(b)** 9 factors; it is a perfect square	count the factor pairs; an odd count signals a repeated pair
6	**(c)** 76.4	$\frac{15 \times 82 + 35 \times 74}{15 + 35} = 76.4$
7	**(a)** 15	groups of equal size, so add the averages and divide by their number
8	**(b)** 10%	$\frac{600 \times 8 + 400 \times 13}{600 + 400} = 10\%$
9	**(a)** 19%	$\frac{80 \times 25 + 120 \times 15}{80 + 120} = 19\%$
10	**(c)** 68	$\frac{70 \times 2 + 80 \times 3 + 60 \times 5}{10} = 68$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If a number is a perfect square, then it has an odd number of factors (2).	2
12	$64 = 2^6$ has factors $1, 2, 4, 8, 16, 32, 64$: 7 factors (1). Odd number of factors, and 64 is a perfect square (8×8) (1).	1+1
13	$\frac{5\times4+4\times3+2\times2}{9} = \frac{36}{9} = \mathbf{4}$ (2).	1+1
14	The total of all the components (1), so the chart helps compare totals while keeping track of the parts (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Checking a few cases gives support only (1). A general claim needs a proof for every n (1). In fact $2^{2^5} + 1 = 2^{32} + 1 = 4294967297 = 641 \times 6700417$ is composite, so the claim fails (1).	1+1+1
16	Equal sizes ⇒ simple average (1½): $\frac{2+3+4+3}{4} = 3$ hours (1½).	1½+1½
17	If a number is the square of a prime, then it has exactly 3 factors (1½). Converse: If a number has exactly 3 factors, then it is the square of a prime (1½).	1½+1½
	OR: (a) Yes, the sticks must be equal since the diagonals are the sticks (1½). (b) Yes, it matters: if the sticks cross at their midpoints a rectangle results; if they cross elsewhere an isosceles trapezium can result, so equal sticks alone do not fix the shape (1½).	1½+1½
18	Profit $= 2400 + 500 = 2900$ (1½). $\frac{2900}{9000} \times 100 \approx 32.2\%$ (1½).	1½+1½
	OR: Males $= x$, females $= 50 - x$: $46x + 36(50 - x) = 40 \times 50$ (1). $10x = 2000 - 1800 = 200$ (1), so $x = 20$ males (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 15: 1, 3, 5, 15; 33: 1, 3, 11, 33 — 4 divisors each (2). (ii) Converse: If n has exactly 4 divisors, then it is a product of two unequal primes (1). False: 8 has divisors 1, 2, 4, 8 (four divisors) but $8 = 2^3$ is not a product of two unequal primes (2).	2+3
	OR: (i) If a quadrilateral is a rectangle, then its diagonals are equal (1). (ii) If a quadrilateral has equal diagonals, then it is a rectangle (1). (iii) An isosceles trapezium with parallel sides 6 cm and 10 cm and legs 5 cm has equal diagonals but no right angle (1½). (iv) A parallelogram: if a parallelogram has equal diagonals, it is a rectangle (1½).	1+1+1½+1½
20	(i) 2015 total 1000: 60%, 10%, 20%, 10% (1½). 2025 total 2000: 40%, 30%, 20%, 10% (2). (ii) Solar's share rose from 10% to 30% and in MW from 100 to 600, a sixfold rise, the fastest growth (1½).	3½+1½

	OR: (i) Salt $= 30$ mL in 900 mL ⇒ 3.33% (2). (ii) Sugar $= 24$ mL in 900 mL ⇒ 2.67% (2). (iii) The other solution dilutes each substance as the total volume increases (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3 factors	p^2 has $1, p, p^2$
2	**(b)** either true or false	definition
3	**(c)** may be true or false	e.g. “if it rains, the road is wet”
4	**(c)** each implies the other	both the proposition and its converse are true
5	**(d)** A is false but R is true	A is false; R is a true fact that in fact contradicts A
6	**(b)** 82	$\frac{82\times1+74\times2+90\times2}{5}=82$
7	**(d)** 81	$\frac{12\times85+33}{13}=81$
8	**(b)** the two groups have different sizes	the larger group should count for more
9	**(c)** 30%	cooling share $=\frac{300}{1000}\times100=30\%$
10	**(d)** A is false but R is true	A is false (totals unknown); R is true

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converse: If n has exactly 3 factors, then n is the square of a prime (1). It is true: three factors means $1, p, p^2$ for a prime p (1). (No counterexample exists.)	1+1
12	Converse: If a number is divisible by both 2 and 3, then it is divisible by 6 (1). Both are true because 2 and 3 have no common factor (1).	1+1
13	Stacked: bar lengths show actual values and totals differ (1). 100% stacked: every bar has the same length and shows only the shares (1).	1+1
14	25% of $800=\frac{25}{100}\times800=200$ units (2).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $64+225=289=17^2$ ✓ right-angled (1½). (ii) $100+576=676$ but $27^2=729$ × (1½).	1½+1½
16	Total $=1000$ MW (1). Shares: coal 60%, solar 10%, wind 20%, hydro 10% (2).	1+2
17	Converse: If a triangle has two equal angles, then it is isosceles (1). It is true (1). Draw the altitude from the third vertex; the two right triangles are congruent by AAS, so the sides opposite the equal angles are equal (1).	1+1+1
	OR: Number theory: If n is a multiple of 8, then it is a multiple of 4; converse false (12) (1½). Geometry: If a quadrilateral is a square, then all angles are equal; converse false (a rectangle) (1½).	1½+1½
18	Except for the first segment, the bars do not start from 0 (1½). So the segment length must be found by comparing end points, which makes estimation harder, though totals are easy to compare (1½).	1½+1½
	OR: (i) Can be inferred; within one bakery 45% > 35% (1½). (ii) Cannot; the totals of the two bakeries are unknown (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i)(a) $225+400=625=25^2$ ✓ right-angled (1½). (b) $121+3600=3721$; $62^2=3844$ × not right-angled (1½). (ii) Build a right triangle with legs a, b; its hypotenuse is $\sqrt{a^2+b^2}=c$; both triangles have the same three sides, so they are congruent by SSS and the given one has a right angle (2).	3+2
	OR: (i) $n=7$: $49+2=51=3\times17$ is not prime (1½). (ii) $n=2$ is prime but $4-1=3$ is not a multiple of 4 (1½). (iii) $n=18$ is divisible by 9, not by 27 (2).	1½+1½+2
20	(i) $\frac{5\times200+3\times280}{8}=\frac{1000+840}{8}=$ ₹230 per kg (3). (ii) $230\times1.2=$ ₹276 per kg (2).	3+2

	OR: (i) Stacked: totals and absolute components can be compared (but later segments do not start at 0); 100% stacked: relative shares can be compared but actual numbers cannot (2½). (ii) Stacked: which family spent the most overall? 100% stacked: which family has the smallest share for healthcare? (2½)	2½+2½

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** If the school is closed, then it is Sunday.	swap the 'if' part and the 'then' part
2	**(b)** 12	it satisfies the 'if' part but not the 'then' part
3	**(b)** true	15 = 3 × 5 with 3 and 5 having no common factor
4	**(b)** not right-angled, since $a^2 + b^2 \neq c^2$	$36 + 64 = 100$ and $11^2 = 121$
5	**(b)** 10 factors; it is not a perfect square	count the factor pairs; an odd count signals a repeated pair
6	**(d)** 65	$\frac{40 \times 60 + 10 \times 85}{40 + 10} = 65$
7	**(c)** 46	groups of equal size, so add the averages and divide by their number
8	**(d)** 14%	$\frac{250 \times 20 + 750 \times 12}{250 + 750} = 14\%$
9	**(a)** 100	$\frac{100 \cdot 60 + 40y}{100 + y} = 50 \Rightarrow y = 100$
10	**(c)** 68	$\frac{45 \times 4 + 60 \times 5 + 90 \times 6}{15} = 68$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Proposition: If a person lives in Jaipur, then the person lives in Rajasthan (1). Converse: If a person lives in Rajasthan, then the person lives in Jaipur — false (a person in Jodhpur) (1).	1+1
12	$9^2 + 40^2 = 81 + 1600 = 1681 = 41^2$ (1). So the triangle is right-angled, with the right angle opposite the 41 cm side (1).	1+1
13	Profit $= 0.20 \times 2000 + 0.30 \times 3000 = 400 + 900 = 1300$ (1); percentage $= \frac{1300}{5000} \times 100 = \mathbf{26\%}$ (1).	1+1
14	15% (1), the midpoint, because equal quantities have equal weights (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Values: 41, 43, 47, 53, 61 are prime (1). Checking a few cases does not prove it (1). $n = 41$ gives $41^2 - 41 + 41 = 1681 = 41 \times 41$, composite (1).	1+1+1
16	Let y litres be added: $\frac{0.05 \times 6 + 0.20y}{6 + y} = 0.08$ (1). $0.3 + 0.2y = 0.48 + 0.08y$ (1), $0.12y = 0.18$, $y = 1.5$ litres (1).	1+1+1
17	Converse: If $a + x = a + y$, then $x = y$ (1). Proposition: adding a to equals gives equals; converse: subtract a from both sides (1). Used while solving equations by balancing (1).	1+1+1
	OR: Converse: If n is not a multiple of 3, then n and $n + 3$ have no common factor (1). Their common factors divide the difference 3, so only 1 or 3 can be common (1). If n is not a multiple of 3, 3 does not divide n, so the converse is true (1).	1+1+1
18	Rashi: $\frac{180 + 280 + 400}{12} = \frac{860}{12} \approx 71.67$ (1). Keerthi: $\frac{210 + 240 + 375}{12} = \frac{825}{12} = 68.75$ (1). Rashi is ahead (1).	1+1+1
	OR: $\frac{4.2 \times 6 + 3.4 \times 5 + 2.8 \times 4}{15} = \frac{25.2 + 17 + 11.2}{15} = \frac{53.4}{15} = 3.56$ (3).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Proposition: equal sides ⇒ equal opposite angles (isosceles triangle theorem); converse: equal angles ⇒ equal opposite sides, proved by the altitude and AAS (2½). (ii) Proposition: "If a number is a multiple of 3, it is a multiple of 5" is false (e.g. 6); its converse "If a number is a multiple of 5, it is a multiple of 3" is also false (e.g. 10) (2½).	2½+2½
	OR: (i) 18=3×6, 24=3×8, 30=3×10 ✓ (1). (ii) If a number is divisible by 3, then it is divisible by 6 (1). (iii) 9, 15, 21 are divisible by 3 but not by 6: counterexamples; 12 is divisible by both (2). (iv) Of 9, 15, 21, 12 only 12 is divisible by 6: $\frac{1}{4}$ (1).	1+1+2+1
20	(i) $\frac{25 + 4 \times 0.5}{5} = \frac{27}{5} = 5.4\%$ (1½). (ii) Yes, 3% lies between 0.5% and 25%: $\frac{25 + 0.5y}{1 + y} = 3 \Rightarrow 25 + 0.5y = 3 + 3y \Rightarrow y = 8.8$ litres (2½). (iii) No; a mixture's concentration always lies between the two concentrations mixed (1).	1½+2½+1

	OR: (i) Option 1: $\frac{80+60+70}{3} = 70$ (1); Option 2: $\frac{80+120+350}{8} = \frac{550}{8} = 68.75$ (1½). (ii) Option 2 gives more weight to the exam; for a student scoring higher in the exam it is better (1½). (iii) Weighted mean (1).	2½+1½+1

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ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4 factors	pq has $1, p, q, pq$
2	**(d)** satisfies the 'if' part but fails the 'then' part	it shows the proposition is false
3	**(d)** composite	$2^{32}+1 = 641 \times 6700417$
4	**(a)** both true	adding 5 can be undone by subtracting 5
5	**(a)** Both A and R are true and R is the correct explanation of A	satisfies 'if', fails 'then'
6	**(d)** 62	$\frac{50\times5+80\times3+65\times2}{10} = 62$
7	**(a)** 3 litres	12×10% = 1.2 litres of solute; total volume = 15 litres
8	**(d)** $\frac{w_1x_1+w_2x_2+\cdots+w_nx_n}{w_1+w_2+\cdots+w_n}$	each value counts by its weight
9	**(b)** 45%	cooling share $= \frac{450}{1000}\times100 = 45\%$
10	**(a)** Both A and R are true and R is the correct explanation of A	equal weights

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	For $n=4$: $2^4+1=17$ is prime, but for $n=6$: $2^6+1=65=5\times13$ is composite (2).	1+1
12	A proposition claims truth for all cases (1). One case where the 'if' part holds but the 'then' part fails contradicts the claim, e.g. 9 for "every odd number is prime" (1).	1+1
13	It tells how important or how big each value is relative to the others (1), so each value is multiplied by its weight before dividing by the total weight (1).	1+1
14	No (1). The chart gives only percentages of each shop's own total; Q's total could be much larger (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a number is a multiple of 2 and of 4, then it is a multiple of 8 (1). Not sufficient (1): 12 is divisible by 2 and 4 but not by 8 (1).	1+1+1
16	Annual marks: weights are the importance of internals, project, exam (1½). Mixtures/alloys: weights are the quantities mixed (1½).	1½+1½
17	Converse: If a number is divisible by 3, then the sum of its digits is a multiple of 3 (1½). It is true: the divisibility rule works in both directions, e.g. 741 and $7+4+1=12$ (1½).	1½+1½
	OR: Counterexample: a 6 cm × 4 cm right triangle and a 12 cm × 2 cm right triangle both have area 12 cm² but are not congruent (1½). Converse: If two triangles are congruent, they have equal area — true (1½).	1½+1½
18	A: $0.20\times7000 = ₹1400$; B: $0.30\times4000 = ₹1200$ (2). Family A spends more rupees although B's percentage is higher (1).	2+1
	OR: Let each quantity be y: concentration $= \frac{\frac{p}{100}y+\frac{q}{100}y}{2y}$ (1½) $= \frac{p+q}{200} = \frac{p+q}{2}\%$ (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The argument starts from 'odd number of factors' and ends at 'perfect square', i.e. the direction (odd factors ⇒ square), the converse (2). (ii) Start from $n=f\times f$; the pair (f,f) exists; two such repeated pairs cannot exist, so all other factors pair up in twos and the count is odd ⇒ proposition proved (3).	2+3
	OR: (i) If my profit is more, then I have sold more articles (1). (ii) 10 pens × ₹4 = ₹40; 20 pens × ₹1 = ₹20; more pens, less profit (1½). (iii) 5 laptops at ₹2000 profit = ₹10000 versus 50 pens at ₹10 profit = ₹500: more profit with fewer articles sold (1½). (iv) Counterexamples (1).	1+1½+1½+1

20	(i) $\frac{2\times3+5\times4+3\times3}{3+4+3}=\frac{6+20+9}{10}=3.5$ m (3). (ii) Simple average $=\frac{2+5+3}{3}=3.33$ m; it ignores the different lengths for which each depth applies (2).	3+2
	OR: (i) $\frac{12\times31+6\times30+2\times31}{31+30+31}$ (2). (ii) $\frac{372+180+62}{92}=\frac{614}{92}\approx6.67$ mm per day (3).	2+3

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